

Overview of ResNet deep learning

Subject: ResNet deep learning

$$y = F(x, \{W_i\}) + x$$

1.

Challenges As with ANNs, many issues can arise with naively trained DNNs. Two common issues are overfitting and computation time. DNNs are prone to overfitting because of the added layers of abstraction, which allow them to model rare dependencies in the training data. Regularization methods such as Ivakhnenko's unit pruning or weight decay (ℓ_2 -regularization) or sparsity (ℓ_1 -regularization) can be applied during training to combat overfitting. Alternatively dropout regularization randomly omits units from the hidden layers during training. This helps to exclude rare dependencies. Another interesting recent development is research into models of just enough complexity through an estimation of the intrinsic complexity of the task being modelled. This approach has been successfully applied for multivariate time series prediction tasks such as traffic prediction. Finally, data can be augmented via methods such as cropping and rotating such that smaller training sets can be increased in size to reduce the chances of overfitting. DNNs must consider many training parameters, such as the size (number of layers and number of units per layer), the learning rate, and initial weights. Sweeping through the parameter space for optimal parameters may not be feasible due to the cost in time and computational resources. Various tricks, such as batching (computing the gradient on several training examples at once rather than individual examples) speed up computation. Large processing capabilities of many-core architectures (such as GPUs or the Intel Xeon Phi) have produced significant speedups in training, because of the suitability of such processing architectures for the matrix and vector computations. Alternatively, engineers may look for other types of neural networks with more straightforward and convergent training algorithms. CMAC (cerebellar model articulation controller) is one such kind of neural network. It doesn't require learning rates or randomized initial weights. The training process can be guaranteed to converge in one step with a new batch of data, and the computational complexity of the training algorithm is linear with respect to the number of neurons involved.

2.

Image reconstruction Image reconstruction is the reconstruction of the underlying images from the image-related measurements. Several works showed the better and superior performance of the deep learning methods compared to analytical methods for various applications, e.g., spectral imaging and ultrasound imaging.

3.

Relation to human cognitive and brain development Deep learning is closely related to a class of theories of brain development (specifically, neocortical development) proposed by cognitive neuroscientists in the early 1990s. These developmental theories were instantiated in computational

models, making them predecessors of deep learning systems. These developmental models share the property that various proposed learning dynamics in the brain (e.g., a wave of nerve growth factor) support the self-organization somewhat analogous to the neural networks utilized in deep learning models. Like the neocortex, neural networks employ a hierarchy of layered filters in which each layer considers information from a prior layer (or the operating environment), and then passes its output (and possibly the original input), to other layers. This process yields a self-organizing stack of transducers, well-tuned to their operating environment. A 1995 description stated, "...the infant's brain seems to organize itself under the influence of waves of so-called trophic-factors ... different regions of the brain become connected sequentially, with one layer of tissue maturing before another and so on until the whole brain is mature". A variety of approaches have been used to investigate the plausibility of deep learning models from a neurobiological perspective. On the one hand, several variants of the backpropagation algorithm have been proposed in order to increase its processing realism. Other researchers have argued that unsupervised forms of deep learning, such as those based on hierarchical generative models and deep belief networks, may be closer to biological reality. In this respect, generative neural network models have been related to neurobiological evidence about sampling-based processing in the cerebral cortex. Although a systematic comparison between the human brain organization and the neuronal encoding in deep networks has not yet been established, several analogies have been reported. For example, the computations performed by deep learning units could be similar to those of actual neurons and neural populations. Similarly, the representations developed by deep learning models are similar to those measured in the primate visual system both at the single-unit and at the population levels.

4.

1980s-2000s The time delay neural network (TDNN) was introduced in 1987 by Alex Waibel to apply CNN to phoneme recognition. It used convolutions, weight sharing, and backpropagation. In 1988, Wei Zhang applied a backpropagation-trained CNN to alphabet recognition. In 1989, Yann LeCun et al. created a CNN called LeNet for recognizing handwritten ZIP codes on mail. Training required 3 days. In 1990, Wei Zhang implemented a CNN on optical computing hardware. In 1991, a CNN was applied to medical image object segmentation and breast cancer detection in mammograms. LeNet-5 (1998), a 7-level CNN by Yann LeCun et al., that classifies digits, was applied by several banks to recognize hand-written numbers on checks digitized in 32x32 pixel images. Recurrent neural networks (RNN) were further developed in the 1980s. Recurrence is used for sequence processing, and when a recurrent network is unrolled, it mathematically resembles a deep feedforward layer. Consequently, they have similar properties and issues, and their developments had mutual influences. In RNN, two early influential works were the Jordan network (1986) and the Elman network (1990), which applied RNN to study problems in cognitive psychology. In the 1980s, backpropagation did not work well for deep learning with long credit assignment paths. To overcome this problem, in 1991, Jürgen Schmidhuber proposed a hierarchy of RNNs pre-trained one level at a time by self-supervised learning where each RNN tries to predict its own next input, which is the next unexpected input of the RNN below. This "neural history compressor" uses predictive coding to learn internal representations at multiple self-organizing time scales. This can substantially facilitate downstream deep learning. The RNN hierarchy can be collapsed into a single RNN, by distilling a higher level chunker network into a lower level automatizer network. In 1993, a neural history compressor solved a "Very Deep Learning" task that required more than 1000 subsequent layers in an RNN unfolded in time. The "P" in ChatGPT refers to such pre-training. Sepp Hochreiter's diploma thesis (1991) implemented the neural history

compressor, and identified and analyzed the vanishing gradient problem. Hochreiter proposed recurrent residual connections to solve the vanishing gradient problem. This led to the long short-term memory (LSTM), published in 1995. LSTM can learn "very deep learning" tasks with long credit assignment paths that require memories of events that happened thousands of discrete time steps before. That LSTM was not yet the modern architecture, which required a "forget gate", introduced in 1999, which became the standard RNN architecture. In 1991, Jürgen Schmidhuber also published adversarial neural networks that contest with each other in the form of a zero-sum game, where one network's gain is the other network's loss. The first network is a generative model that models a probability distribution over output patterns. The second network learns by gradient descent to predict the reactions of the environment to these patterns. This was called "artificial curiosity". In 2014, this principle was used in generative adversarial networks (GANs). During 1985–1995, inspired by statistical mechanics, several architectures and methods were developed by Terry Sejnowski, Peter Dayan, Geoffrey Hinton, etc., including the Boltzmann machine, restricted Boltzmann machine, Helmholtz machine, and the wake-sleep algorithm. These were designed for unsupervised learning of deep generative models. However, those were more computationally expensive compared to backpropagation. Boltzmann machine learning algorithm, published in 1985, was briefly popular before being eclipsed by the backpropagation algorithm in 1986. (p. 112). A 1988 network became state of the art in protein structure prediction, an early application of deep learning to bioinformatics. Both shallow and deep learning (e.g., recurrent nets) of ANNs for speech recognition have been explored for many years. These methods never outperformed non-uniform internal-handcrafting Gaussian mixture model/Hidden Markov model (GMM-HMM) technology based on generative models of speech trained discriminatively. Key difficulties have been analyzed, including gradient diminishing and weak temporal correlation structure in neural predictive models. Additional difficulties were the lack of training data and limited computing power. Most speech recognition researchers moved away from neural nets to pursue generative modeling. An exception was at SRI International in the late 1990s. Funded by the US government's NSA and DARPA, SRI researched in speech and speaker recognition. The speaker recognition team led by Larry Heck reported significant success with deep neural networks in speech processing in the 1998 NIST Speaker Recognition benchmark. It was deployed in the Nuance Verifier, representing the first major industrial application of deep learning. The principle of elevating "raw" features over hand-crafted optimization was first explored successfully in the architecture of deep autoencoder on the "raw" spectrogram or linear filter-bank features in the late 1990s, showing its superiority over the Mel-Cepstral features that contain stages of fixed transformation from spectrograms. The raw features of speech, waveforms, later produced excellent larger-scale results.